Debugging and optimization

1

```
import java.util.Scanner;

public class Debug1 {
public static void main(String args[]) {
Scanner sc = new Scanner(System.in);

int arr[] = {20,40,60,80};

int index = sc.nextInt();
int divisor = sc.nextInt();

try {
int value = arr[index];
try {
int result = value/divisor;
System.out.println(result);
}
catch(ArrayIndexOutOfBoundsException e) {
System.out.println("Invalid Index");
}
}
catch(ArithmeticException e) {
System.out.println("Division by Zero");
}
}
}
```

```
import java.util.Scanner;
public class Debug1 {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        int arr[] = {20, 40, 60, 80};
        if (sc.hasNextInt()) {
            int index = sc.nextInt();
            if (sc.hasNextInt()) {
                int divisor = sc.nextInt();
                try {
                    int value = arr[index];
                    try {
                        int result = value / divisor;
                        System.out.println(result);
                    } catch (ArithmeticException e) {
                        System.out.println("Division by Zero");
                    }
                } catch (ArrayIndexOutOfBoundsException e) {
                    System.out.println("Invalid Index");
                }
            }
        }
        sc.close();
    }
}
```

```
1
2
```

Output

```
20
```

Test Cases

- Test Case 1 ✔
- Test Case 2 ✔
- Test Case 3 ✔

10.00 / 10 marks

2

```
import java.util.Scanner;

class Calculator{

static int divide(int a,int b){
return a/b;
}

static void display(int x){
System.out.println("Answer = "+x);
}

}

public class Debug2{

public static void main(String args[]){

Scanner sc=new Scanner(System.in);

int a=sc.nextInt();
int b=sc.nextInt();

Calculator.display(Calculator.divide(a,b));

System.out.println("Execution Completed");
}

}
```

Debugging Task

Prevent abnormal termination.

}

Debugging Task

Prevent abnormal termination.

Ensure "Execution Completed" is always printed.

TEST CASE 1

INPUT: 20 5

OUTPUT : Answer = 4

Execution Completed

TEST CASE 2

INPUT: 50 0

OUTPUT: Cannot Divide by Zero

Execution Completed

TEST CASE 3

INPUT: 36 6

OUTPUT: Answer = 6

Execution Completed

Your Fixed Code

```java
import java.util.Scanner;
public class Debug2 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        try {
            int a = sc.nextInt();
            int b = sc.nextInt();
            Calculator.display(Calculator.divide(a, b));
        } catch (ArithmeticException e) {
            System.out.println("Cannot Divide by Zero");
        } catch (Exception e) {
        } finally {
            System.out.println("Execution Completed");
            sc.close();
        }
    }
}
class Calculator {
    static int divide(int a, int b) {
        return a / b;
    }
    static void display(int x) {
        System.out.println("Answer = " + x);
    }
}
```

Test Input

```
20 5
```

Output

```
Answer = 4
Execution Completed
```

Test Cases

- Test Case 1 ✔
- Test Case 2 ✔
- Test Case 3 ✔

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3

```java
import java.util.Scanner;

public class Debug3 {

public static void main(String args[]) {

Scanner sc=new Scanner(System.in);

int arr[]={12,24,36,48};

String number=sc.next();

int index=Integer.parseInt(number);

System.out.println(arr[index]);

}
}
```

Debugging Task

The program may generate two different runtime exceptions.

Modify the program so that each exception is handled separately.

Test Case 1

INPUT: 2

OUTPUT: 36

Test Case 2

INPUT: ABC

OUTPUT: Invalid Number

Test Case 3

INPUT: 5

OUTPUT: Invalid Array Index

ENG

Your Fixed Code

```java
import java.util.Scanner;
public class Debug3 {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        int arr[] = {12, 24, 36, 48};
        String number = sc.next();
        try {
            int index = Integer.parseInt(number);
            try {
                System.out.println(arr[index]);
            } catch (ArrayIndexOutOfBoundsException e) {
                System.out.println("Invalid Array Index");
            }
        } catch (NumberFormatException e) {
            System.out.println("Invalid Number");
        }
    }
}
```

Test Input

```
2
```

Output

```
36
```

Test Cases

Test Case 1 ✔

Test Case 2 ✔

Test Case 3 ✔

10.00 / 10 marks

Submitted

4

```java
public class Debug4 {

static int calculate(int a,int b){

try{
return a/b;
}
finally{
System.out.println("Resources Released");
}

}

public static void main(String args[]){

System.out.println(calculate(20,0));

}

}
```

Debugging Task

Prevent program termination.

Ensure the finally block executes while handling the exception correctly.

Test Case 1

INPUT: 20 5

OUTPUT: Resources Released

4

Test Case 2

20 0

Ensure the finally block executes while handling the exception correctly.

Test Case 1

INPUT: 20 5

OUTPUT: Resources Released

4

Test Case 2

20 0

Resources Released

Division by Zero

Test Case 3

81 9

Resources Released

9

Your Fixed Code

```
public class Debug4 {
    static int calculate(int a, int b) throws ArithmeticException {
        try {
            return a / b;
        } finally {
            System.out.println("Resources Released");
        }
    }
    public static void main(String args[]) {
        java.util.Scanner sc = new java.util.Scanner(System.in);
        if (sc.hasNextInt()) {
            int a = sc.nextInt();
            int b = sc.nextInt();
            try {
                int result = calculate(a, b);
                System.out.println(result);
            } catch (ArithmeticException e) {
                System.out.println("Division by Zero");
            }
        }
        sc.close();
    }
}
```

Test Input

```
20 5
```

Output

```
Resources Released
4
```

Test Cases

- Test Case 1
- Test Case 2
- Test Case 3

10.00 / 10 marks

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```java
public class Debug4 {
public static void main(String[] args) {
String name = null;
System.out.println(name.length());
}
}
```

Expected Task

Prevent the program from crashing.

Test Case 1

INPUT: Java

OUTPUT: Length = 4

Test Case 2

INPUT: null

OUTPUT: String is Null

Test Case 3

INPUT: Programming

OUTPUT: Length = 11

Your Fixed Code

```java
import java.util.Scanner;
public class Debug4 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        if(!sc.hasNext()){
            System.out.println("String is null");
            return;
        }
        String s=sc.nextLine();
        if (s.equals("null")) {
            System.out.println("String is Null");
        } else {
            System.out.println("Length = "+s.length());
        }
    }
}
```

Test Input

```
java
```

Output

```
Compilation Error:
```

Test Cases

- Test Case 1
- Test Case 2
- Test Case 3

10.00 / 10 marks